

PINCHER CREEK, AB, Canada

WMO: 710850

Lat: 49.5222N

Lon: 114.0050W

Elev: 1190

StdP: 87.83

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-28.6	-25.0	-32.9	0.2	-28.5	-29.1	0.3	-24.8	21.6	1.0	19.4	0.8	3.1	280	0.818

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.4	29.5	15.5	27.6	14.9	25.8	14.4	17.4	25.6	16.4	24.5	15.4	23.5	6.3	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.5	11.9	20.4	13.3	11.0	19.2	12.2	10.2	18.0	53.6	25.6	50.3	24.6	47.1	23.6	22.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.0	15.8	13.7	DB	-32.9	32.3	3.6	2.1	-35.5	33.8	-37.6	35.1	-39.6	36.2	-42.3	37.8
			WB	-33.0	19.7	3.6	1.3	-35.6	20.6	-37.7	21.3	-39.7	22.1	-42.3	23.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.1	-5.1	-4.7	-1.2	4.1	9.1	13.0	17.0	16.4	11.9	5.6	-0.7	-5.3
	DBStd	10.11	8.76	8.14	7.73	5.24	4.38	3.25	3.05	3.27	4.64	5.44	7.38	8.23
	HDD10.0	2473	468	412	347	184	69	9	1	2	32	153	322	473
	HDD18.3	4883	727	645	604	427	286	161	62	77	196	396	571	731
	CDD10.0	674	0	0	1	7	42	100	217	199	90	16	2	0
	CDD18.3	41	0	0	0	0	1	2	19	16	3	0	0	0
	CDH23.3	1013	0	0	0	1	26	61	420	392	112	2	0	0
	CDH26.7	259	0	0	0	0	2	11	121	106	20	0	0	0
(14) Wind	WSAvg	6.1	7.5	6.5	6.3	5.8	5.5	5.4	4.8	4.6	5.3	6.5	7.1	7.4
(15) Precipitation	PrecAvg	696	51	42	50	51	79	97	49	46	51	51	63	53
	PrecMax	924	79	72	98	95	153	258	167	134	153	89	139	116
	PrecMin	510	16	12	17	29	14	38	8	9	6	20	16	24
	PrecStd	126	17	18	21	16	37	50	39	28	33	24	35	21
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.0	11.4	15.5	21.3	26.1	28.4	32.2	31.8	29.2	21.6	15.4	9.7
		MCWB	4.8	4.4	6.4	9.9	13.2	16.7	16.6	15.6	14.0	11.0	7.5	4.4
	2%	DB	7.9	8.7	12.3	18.2	23.4	24.9	29.8	29.4	26.5	18.3	12.2	7.2
		MCWB	2.8	3.4	5.5	8.7	12.0	14.5	16.1	15.0	13.2	9.5	6.4	2.7
	5%	DB	6.2	6.6	10.1	15.3	20.7	22.7	27.7	27.5	24.0	15.7	9.9	5.5
		MCWB	2.1	2.1	4.3	7.0	10.8	13.8	15.5	14.7	12.7	8.4	4.9	1.7
	10%	DB	4.7	4.9	8.1	12.5	18.0	20.5	25.7	25.5	21.1	13.4	8.0	4.1
		MCWB	1.3	1.0	3.1	5.6	9.6	12.5	15.3	14.2	11.5	7.1	3.7	0.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.5	5.2	7.5	10.6	14.2	17.9	19.4	17.9	15.4	12.0	8.8	5.1
		MCDB	10.0	10.2	13.9	19.8	23.9	26.2	27.2	26.5	24.8	19.9	13.8	8.8
	2%	WB	3.5	3.8	6.0	8.9	12.6	15.7	17.9	16.6	14.0	10.2	6.9	3.3
		MCDB	7.0	7.9	11.5	17.1	21.8	23.1	26.0	25.4	23.9	17.1	11.8	6.5
	5%	WB	2.4	2.5	4.6	7.4	11.4	14.4	16.9	15.7	13.0	8.8	5.2	2.0
		MCDB	5.6	6.4	9.6	14.6	19.7	21.3	25.0	24.5	22.9	15.2	9.5	5.1
	10%	WB	1.3	1.1	3.3	6.0	10.1	13.1	16.0	14.8	11.9	7.3	3.8	0.7
		MCDB	4.7	4.7	7.8	12.1	17.2	19.2	23.9	23.6	20.4	12.7	7.8	4.0
(35) Mean Daily Temperature Range	5% DB	MDBR	9.0	9.6	9.9	10.9	11.8	11.6	14.4	14.9	13.0	10.1	8.3	8.2
		MCDBR	8.4	10.1	11.9	15.8	17.1	16.2	18.7	19.4	18.8	13.4	9.5	7.6
		MCWBR	5.8	6.5	7.1	8.5	8.3	8.3	8.0	7.9	8.3	7.1	5.9	5.5
	5% WB	MCDBR	8.3	10.3	11.5	14.9	15.7	14.8	16.3	16.8	17.2	12.9	9.7	7.8
		MCWBR	6.2	7.1	7.2	8.2	7.9	8.0	8.2	7.6	8.0	7.1	6.4	6.1
(40) Clear-Sky Solar Irradiance	taub		0.243	0.254	0.282	0.339	0.344	0.337	0.336	0.350	0.315	0.282	0.259	0.232
	taud		2.396	2.390	2.414	2.293	2.374	2.435	2.449	2.400	2.476	2.549	2.496	2.419
	Ebn at Noon		837	913	938	906	914	921	916	886	889	871	815	807
	Edh at Noon		62	80	94	119	115	109	106	106	88	69	56	52
(44) All-Sky Solar Radiation	RadAvg		1.17	2.15	3.30	4.66	5.55	6.10	6.92	5.59	3.92	2.39	1.27	0.92
	RadStd		0.10	0.17	0.29	0.35	0.47	0.47	0.45	0.38	0.37	0.25	0.11	0.09

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (57 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page